Project: Weather Data Analysis

1) Project Summary

Analyze a historical weather dataset to clean, explore, visualize, and produce business-ready insights and a dashboard that help non-technical stakeholders understand climate patterns.

2) Objectives

- Clean and prepare the dataset so it is analysis-ready.
- Perform EDA (summary stats, trends, seasonality).
- Produce clear visualizations and written insights (hottest/coldest days, rainfall patterns).
- Deliver a reproducible Jupyter notebook, a short PDF report, and a simple dashboard (Power BI or Jupyter HTML).
- (Optional) Build a simple forecasting model for temperature or rainfall.

3) Dataset (provided / expected)

Columns: Date (YYYY-MM-DD), Temperature (°C), Humidity (%), WindSpeed (km/h), Rainfall (mm)
Students should use the provided CSV data/weather_data.csv (or gather similar open data). Dataset size: 1,000–10,000 rows recommended.

4) Tools & Environment

- Python 3.8+; Jupyter Notebook/Lab
- Libraries: pandas, numpy, matplotlib, seaborn, scipy (optional), scikit-learn (optional)
- Git & GitHub (or GitLab) for version control
- Power BI or Tableau (optional) for dashboard
- requirements.txt or environment.yml must be included

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5) Deliverables (final)

Place these in repository root:

1. notebooks/Weather_Data_Analysis.ipynb — full annotated notebook (must run start→finish).
2. data/weather_data.csv — dataset used (original or cleaned).
3. reports/Weather_Analysis_Report.pdf — 2–4 page executive summary (visuals + 5 key insights + action items).
4. dashboard/ — Power BI .pbix or exported HTML dashboard + screenshots.
5. README.md — how to run, environment, contacts.
6. requirements.txt or environment.yml
7. plots/ — exported PNGs of main figures.
8. submission.txt — group members, role, time spent, known issues.

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6) Folder structure (exact)

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├── README.md
├── requirements.txt
├── submission.txt
|
├── data/
|   └── weather_data.csv
|
├── notebooks/
|   └── Weather_Data_Analysis.ipynb
|
├── reports/
|   └── Weather_Analysis_Report.pdf
|
├── dashboard/
|   └── dashboard.pbix (or dashboard.html)
|
└── plots/
    ├── temp_trend.png
    ├── rainfall_patterns.png
    └── [other visualizations...]
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7) Step-by-step tasks (phase-wise, company-style with acceptance criteria)

Phase A — Setup & Initial Review

1. **Clone repo / create env**
    - git clone <repo-url>
    - create venv/conda, pip install -r requirements.txt
2. **Open notebook and run initial cells**

- Acceptance: Notebook runs first 5 cells without error and loads weather_data.csv.

Phase B — Data Understanding & Cleaning

1. Load data; inspect df.info(), df.describe(), df.head().
2. Convert Date → datetime, sort by date, set index if needed.
3. Identify & handle missing values (drop/impute), detect outliers, correct types.
4. Create a data cleaning log (in notebook) describing every transformation.
    - Acceptance: Cleaned file data/weather_data_cleaned.csv saved; notebook shows df.isna().sum() = 0 or documented imputation.

Phase C — Exploratory Data Analysis (EDA)

1. Compute summary statistics (mean, median, std) per month/season.
2. Find extremes: hottest day(s), coldest day(s), wettest day(s).
3. Correlation analysis (e.g., Temperature vs Humidity).
    - Acceptance: Notebook contains labeled tables & at least 6 charts (trend, histogram, boxplot, heatmap, monthly means, rolling average).

Phase D — Visualizations & Insights

1. Create publication-quality figures; export PNGs to plots/.
2. Write 5–8 actionable insights in plain language (for stakeholders).
    - Acceptance: reports/Weather_Analysis_Report.pdf includes visuals and a one-paragraph recommendation section.

Phase E — (Optional) Modeling / Forecasting

1. If attempted: split data, build a baseline model (e.g., ARIMA or simple linear regression for seasonality), evaluate with RMSE.
    - Acceptance: Notebook cell with model performance metrics and a short interpretation.

Phase F — Dashboard & Presentation

1. Create a simple dashboard that shows: current trends, month-wise averages, and hotspots.
2. Export PBIX or HTML; include a one-page PPT slide for presentation.
    - Acceptance: Dashboard file present and a 5-minute presentation slide deck.

Phase G — Finalize & Submit

1. Tag release: git tag v1-final
2. Ensure README includes run steps and dataset source.
    - Acceptance: Repo builds on a fresh clone with pip install -r requirements.txt and jupyter nbconvert --to html notebooks/Weather_Data_Analysis.ipynb completes.

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8) Submission instructions

- **Repository name**: WG_<yourgroup>_WeatherProject
- **Final branch**: submission (push one commit, tag v1-final)
- **File naming**: Lastname_Firstname_Roll_Project.zip (only if instructor prefers zipped submission)
- **Deadline**: (instructor to set) — include timestamp in submission.txt

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9) Grading rubric (company-style)

- Data cleaning & reproducibility — **20%**
- EDA & visualizations quality — **25%**
- Insights & business interpretation — **20%**
- Code quality & documentation (README, comments) — **15%**
- Dashboard & presentation — **10%**
- Timely submission & teamwork notes — **10%**

Each item should include short comments from graders (what's missing / how to improve).

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10) Instructor checklist (quick)

- Notebook runs start→finish without errors
- requirements.txt present and installs
- Cleaning log (explain imputations/outliers)
- ≥6 clear visuals with captions
- Executive report (2–4 pages) with 5 key insights
- Dashboard included + slide deck
- Git history present (meaningful commits)
- Submission file naming correct
